MTH401 QUIZ(2)

Lecture: not mentioned

RIZ MUGHAL
SQA ENGINEER:

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MTH401:Quiz NO.02 Quiz Start Time: 09:51 AM, 14 Jun

Question # 1 of 10 (Start time: 09:51:54 AM, 14 June 2021) Total M

The auxiliary equation of the differential equation, $y''' + 2y'' - 3y = 0$, is _____.

Select the correct option

Reload Math Equa

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- $m^3 - 3 = 0$
- $m^3 + 2m^2 = 0$
- $(m^3 + 2m^2 - 3)\, e^{mx} = 0$
- $m^3 + 2m^2 - 3 = 0$

MTH401:Quiz NO.02 Quiz Start Time: 09:51 AM, 14 June 2021

Question # 2 of 10 (Start time: 09:52:17 AM, 14 June 2021) Total Marks: 1

A function satisfying the differential equation on the interval I whose graph passes through (x_0, y_0) such that the slope of the curve at the point is the number y'_0 is called solution of the _____ value problem.

Select the correct option

Reload Math Equations

- (●) initial
- ( ) boundary

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MTH401:Quiz NO.02 Quiz Start Time: 09:5

Question # 3 of 10 (Start time: 09:52:36 AM, 14 June 2021)

The gravitational force exerted by the earth on a body of mass m is called ---------- of the body.

Select the correct option

- Force
- weight

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MTH401:Quiz NO.02 Quiz Start Time: 09:51 AM, 14 June 2021

Question # 4 of 10 (Start time: 09:52:53 AM, 14 June 2021) Total Marks: 1

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = -2$, $W_2 = \frac{e^x}{x}$, then $u'_2 =$ _____.

Select the correct option

Reload Math Equations

- $\frac{e^x}{x}$
- $-\frac{e^x}{2x}$
- $2x\,e^x$
- $\frac{e^{-x}}{2x}$

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MTH401:Quiz NO.02 Quiz Start Time: 09:

Question # 5 of 10 (Start time: 09:53:10 AM, 14 June 2021)

What is annihilator operator of the function

$$g(x) = x^4 + 5$$

?

Select the correct option

Relo

- $(D+5)^4$
- (D-4)
- (D+4)
- D^5

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MTH401:Quiz NO.02

Quiz Start Time: 09:51 AM, 14 June 2021

Question # 6 of 10 (Start time: 09:53:25 AM, 14 June 2021) **Total Marks:** 1

The combination of the Newton's second law and the Hook's law could lead to a differential equation governing to the motion of a mass attached to spring i.e. _____ motion.

Select the correct option

- (●) simple harmonic
- ( ) linear
- ( ) translational
- ( ) rotational

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MTH401:Quiz NO.02 Quiz Start Time: 09:5

Question # 7 of 10 (Start time: 09:53:42 AM, 14 June 2021)

If 3,3 are real roots of a differential equation, then the general solution is_____.

Select the correct option

Reloa

- $y = (c_1 + c_2 x)e^{3x}$
- $y = c_1 e^{3x} - c_2 e^{3x}$
- $y = c_1 e^{3x} + c_2 e^{3x}$
- $y = c_1 e^{3x}$

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MTH401:Quiz NO.02

Quiz Start Time: 09:51 AM, 14 June 2021

Question # 8 of 10 (Start time: 09:53:58 AM, 14 June 2021) **Total Marks:** 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \cos 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

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- $A\cos 4x + B\sin 4x$
- $(A + B)(\cos 4x + \sin 4x)$
- Ae^{4x}
- $Ax^2 + Bx + C$

MTH401:Quiz NO.02 Qu

Question # 9 of 10 (Start time: 09:54:16 AM, 14 June 2021)

The differential operator D possesses the _____ property.

Select the correct option

- inverse
- closure
- linearity
- nonlinearity

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MTH401:Quiz NO.02

Quiz Start Time: 09:51 AM, 14 June 2021

Question # 10 of 10 (Start time: 09:54:33 AM, 14 June 2021)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$. *If* the input function, $g(x) = e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

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- $(Ax^2 + Bx + C)e^{5x}$
- Ae^{5x}
- $Ae^{5x}\cos 5x + Be^{5x}\sin 5x$
- $(Ax + B)e^{5x}$

2nd account

MTH401:Quiz NO.02 Quiz Start Time: 10:19 AM, 1

Question # 1 of 10 (Start time: 10:19:49 AM, 14 June 2021) T

If 2,3,5 are real roots of a differential equation, then the general solution is_____.

Select the correct option

Reload Math

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- $y_c = \[(c_1 + c_2 + c_3)e^{3x} + e^{2x}$
- $y_c = c_1e^{2x} + c_2e^{-3x} + c_3e^{5x}$
- $y_c = c_1e^{2x} + c_2e^{3x} + c_3e^{5x}$
- $y_c = c_1e^{2x} + c_1e^{-3x} + c_1e^{5x}$

MTH401:Quiz NO.02 Quiz Start Time: 10:19 AM, 14 June 202

Question # 2 of 10 (Start time: 10:20:20 AM, 14 June 2021) Total Marks:

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$. *If* the input function, $g(x) = e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

- Ae^{5x}
- $(Ax^2 + Bx + C)e^{5x}$
- $(Ax + B)e^{5x}$
- $Ae^{5x}\cos 5x + Be^{5x}\sin 5x$

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MTH401:Quiz NO.02 Quiz Start

Question # 3 of 10 (Start time: 10:20:38 AM, 14 June 2021)

$$Wronskian\ of\ 1\ and\ x^2,\ W(1, x^2) = _____$$

Select the correct option

- x
- 2x
- 0
- None of these

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MTH401:Quiz NO.02

Quiz Start Time: 10:19 AM, 14 June 2021

Question # 4 of 10 (Start time: 10:20:53 AM, 14 June 2021)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$. *If* the input function, $g(x) = x^3 - x + 1$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

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- $Ax^2 + Bx + C$
- $Ax + B$
- A
- $Ax^3 + Bx^2 + Cx + D$

MTH401:Quiz NO.02

Quiz Start Time: 10:19 AM, 14 June 20

Question # 5 of 10 (Start time: 10:21:10 AM, 14 June 2021)

Total Mark

The combination of the Newton's second law and the Hook's law could lead to a differential equation governing to the motion of a mass attached to spring i.e. _____ motion.

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Select the correct option

- rotational
- simple harmonic
- translational
- linear

MTH401:Quiz NO.02 Quiz Start Time: 10:19 AM, 14 June 202

Question # 6 of 10 (Start time: 10:21:28 AM, 14 June 2021) Total Marks:

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 2$, $W_1 = \frac{e^{-x}}{x}$, then $u'_1 =$ _____.

Select the correct option

Reload Math Equations

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- $-\frac{e^{-x}}{2x}$
- $\frac{e^{-x}}{x}$
- $2x\,e^{-x}$
- $\frac{e^{-x}}{2x}$ (selected)

Question # 7 of 10 (Start time: 10:21:48 AM, 14 June 2021)

$$y = e^{(1/2)x}(c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x)$$

is the complementary solution of

$$y'' - y' + y = 2 \sin 3x$$

, then the general form of its particular solution is

$$y_p$$

=_____.

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Select the correct option

Reload

○	$3Asin3x + 3B \cos 3x$
○	$3 \cos 3x + 3 \sin 3x$
○	$3A \cos 3x + 3A \sin 3x$
●	$A \cos 3x + B \sin 3x$

MTH401:Quiz NO.02 Quiz Start Time: 1(

Question # 8 of 10 (Start time: 10:22:11 AM, 14 June 2021)

Wronskian, $W(x, -3x) =$ _____.

Select the correct option

Rel

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- (●) 0
- ( ) 1
- ( ) $3x$
- ( ) $-3x$

MTH401:Quiz NO.02 Quiz Start Time: 10:19 AM

Question # 9 of 10 (Start time: 10:22:29 AM, 14 June 2021)

A set of functions

$$\{f_1(x), f_2(x), f_3(x), \ldots\ldots, f_n(x)\}$$

is said to be _____ on an interval I if there exist constants

$$c_1, c_2, c_3, \ldots\ldots, c_n$$

not all zero, such that

$$c_1 f_1(x) + c_2 f_2(x), \ldots\ldots, c_n f_n(x) = 0, \quad \forall x \in I$$

Select the correct option

Reload Ma

- (●) linear dependent
- ( ) linear independent

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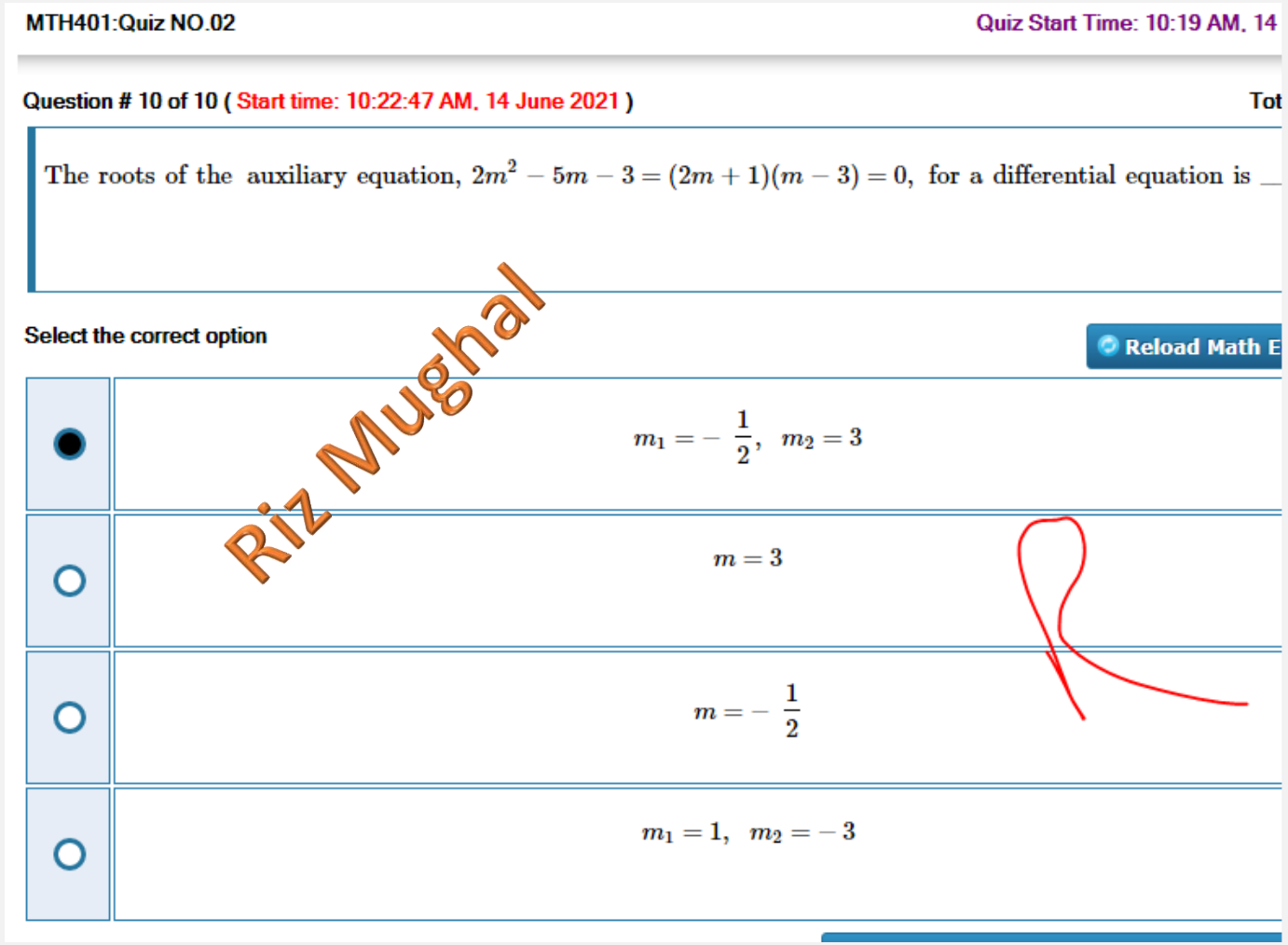
MTH401:Quiz NO.02
Quiz Start Time: 10:19 AM, 14
Question # 10 of 10 (Start time: 10:22:47 AM, 14 June 2021)
Tot
The roots of the auxiliary equation, $2m^2 - 5m - 3 = (2m+1)(m-3) = 0$, for a differential equation is
Select the correct option
Reload Math E
$m_1 = -\frac{1}{2}, \quad m_2 = 3$
$m = 3$
$m = -\frac{1}{2}$
$m_1 = 1, \quad m_2 = -3$
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3rd account

MTH401:Quiz NO.02 Quiz Start Time: (

Question # 1 of 10 (Start time: 03:41:35 PM, 14 June 2021)

What is annihilator operator of the function

$$g(x) = x^4 + 5$$

?

Select the correct option

Re

- D^5
- (D-4)
- $(D+5)^4$
- (D+4)

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MTH401:Quiz NO.02 Quiz Start Time: 03:41

Question # 2 of 10 (Start time: 03:42:07 PM, 14 June 2021)

The periodic time is given by

Select the correct option

Reload

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- $\frac{\omega}{2\pi}$
- $2\pi * \omega$
- $\frac{2\pi}{\omega}$
- $\frac{\pi}{\omega}$

MTH401:Quiz NO.02 Quiz Start Tin

Question # 3 of 10 (Start time: 03:42:27 PM, 14 June 2021)

The differential operator D transforms a _____ function into another function.

Select the correct option

- integrable
- differentiable
- discontinuous
- continuous

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MTH401:Quiz NO.02 Quiz Start

Question # 4 of 10 (Start time: 03:42:57 PM, 14 June 2021)

How are frequency and period related in simple harmonic motion?

Select the correct option

- None of the above
- Their sum is constant
- They are inversely related
- They are directly related

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MTH401:Quiz NO.02 Quiz Start Time:

Question # 5 of 10 (Start time: 03:43:16 PM, 14 June 2021)

$$Wronskian\ of\ 1\ and\ x^2,\ W(1, x^2) = _____$$

Select the correct option

- (●) 2x
- ( ) x
- ( ) None of these
- ( ) 0

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MTH401:Quiz NO.02

Quiz Start Time: 03:41 PM, 14 June 20

Question # 6 of 10 (Start time: 03:43:36 PM, 14 June 2021) **Total Marks**

We can derive a differential equation governing the motion of a mass attached to spring when the Newton's second law combined with _____.

Select the correct option

- (●) Hook's law
- ( ) Newton's 3rd law

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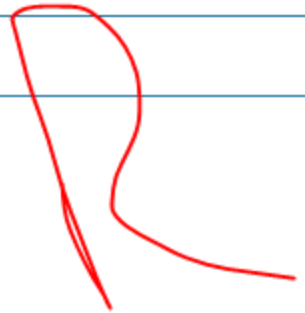

MTH401:Quiz NO.02

Quiz Start Time: 03:41 PM, 14 June 2(

Question # 7 of 10 (Start time: 03:43:56 PM, 14 June 2021)

Total Mark

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$. *If* the input function, $g(x) =$ constant (say 1), then the assumed particular solution y_p could be _____

Select the correct option

Reload Math Equatio

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- $Ax + B$
- $Ax^2 + Bx + C$
- $Ax^3 + Bx^2 + Cx + D$
- A

MTH401:Quiz NO.02 Quiz Start Time: 03:41 PM, 14 J

Question # 8 of 10 (Start time: 03:44:16 PM, 14 June 2021) Tota

The roots of the auxiliary equation, $m^2 - 10m + 25 = (m-5)^2 = 0$, for a differential equation is ___

Select the correct option

Reload Math Eq

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- $m = 25$
- $m_1 = m_2 = 25$
- $m = -10$
- $m_1 = m_2 = 5$

MTH401:Quiz NO.02 Quiz Start Time: 03:41 PM, 14 June 2021

Question # 9 of 10 (Start time: 03:44:38 PM, 14 June 2021) Total Marks: 1

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 2$, $W_1 = \frac{e^{-x}}{x}$, then $u'_1 =$ _____.

Select the correct option

Reload Math Equations

- $\frac{e^{-x}}{2x}$
- $-\frac{e^{-x}}{2x}$
- $\frac{e^{-x}}{x}$
- $2x\,e^{-x}$

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MTH401:Quiz NO.02 Quiz Start Time: 03:4

Question # 10 of 10 (Start time: 03:44:59 PM, 14 June 2021)

The gravitational force exerted by the earth on a body of mass m is called ---------- of the body.

Select the correct option

- (●) weight
- ( ) Force

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Thank you for watching ☺

Share with your fellows

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